RANDOM FOREST MACHINE LEARNING PROJECT

Pima Indians Diabetes Dataset | Classification

1. Problem Statement

The objective is to build a classification model to predict whether a person has diabetes (Outcome = 1) or does not have diabetes (Outcome = 0) using health-related features.

2. Dataset

Pima Indians Diabetes Database, obtained from Kaggle. The dataset contains 768 records and 9 columns: eight input features and the target column, Outcome.

3. Data Preprocessing

• Added column headings because the CSV had no header row.
• Explored the dataset and checked missing values and duplicates.
• Treated zero measurements in Glucose, BloodPressure, SkinThickness, Insulin and BMI as missing, then imputed them using each column's median.
• Applied IQR-based outlier capping to numerical input features, excluding Outcome.
• Separated X and y and performed an 80:20 stratified train-test split.

4. Model Used

RandomForestClassifier from scikit-learn with 100 decision trees (n_estimators=100) and random_state=42. The model was trained on the training set and evaluated on the test set.

5. Evaluation Results

Metric	Result
Accuracy	0.7468 (74.68%)
Precision – class 1	0.6667 (66.67%)
Recall – class 1	0.5556 (55.56%)
F1-score – class 1	0.6061 (60.61%)

The classification report also showed class 0 precision = 0.78, recall = 0.85 and F1-score = 0.81; class 1 precision = 0.67, recall = 0.56 and F1-score = 0.61.

6. Key Findings

The model achieved approximately 75% accuracy on the test set. Recall for the diabetes-positive class (class 1) was 0.56, meaning some actual positive cases were missed. The feature-importance ranking was not provided, so it should be added from the model's feature_importances_ output rather than assumed.

7. Conclusion

A Random Forest model was developed to predict diabetes outcomes using the Pima Indians Diabetes dataset. The project covered data exploration, missing-value treatment, IQR outlier capping, train-test splitting, model training and evaluation. The model achieved 74.68% test accuracy. Hyperparameter tuning and cross-validation could be explored to improve performance. This is an educational project, not a medical diagnostic tool.

Dataset source: Kaggle – Pima Indians Diabetes Database (https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database)